

PV Feasibility Study: Buyer and Bellway Benefit Analysis

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June 2026

0.1 Executive Summary

Bellway's current building fabric and ASHP already meet Scottish Building regulation under section 6 energy requirements without PV. However PV panels are widely perceived by buyers as a visual identifier for efficient green energy and may offer a marketing value beyond regulatory compliance. This report assesses the financial and marketing case for installing PV across four Bellway house types using a hypothetical combination of real PV generation data Blindwells and real heat pump data Kelty since no site has current data for both.

0.2 Methodology and key assumptions

No single Bellway site currently combines both ASHP and PV as standard. PV generation data was therefore taken from Blindwells (east Lothian), a live Bellway site with PV installation but gas heating. Whereas ASHP and SAP energy performance data was taken from Kelty, a Bellway site with ASHP as a standard but no PV. This report combines both sets of data for a hypothetical scenario for each of the four house types.

- House Types modeled: kinloch (3 bed, $86m^2$), Rosedale(3 Bed $110m^2$), Pinehurst(4 bed $128m^2$), Sunningdale (5bed, $167m^2$)
- PV system: Viridian Pv16-335G1, 335W monocrystalline, 20.7% efficiency
- ASHP: Grant logic air mono block, COP at A2/W30 (4.66/4.29/4 unit size)
- Mandatory 7.4KW Datec eDock2 EVcharger ; Ev demand assumed 214KWh/yr (7400mi/yr/3.5mi/kWh, UK average)
- electricity unit rate: 24.4p/kWh; SEG export rates:12p average(no battery), 29p (octopus flux,with battery)
- Solar irradiance: $1050 KWh/m^2$ /yr (East Lothian)
- Performance ratio :0.8 (standard for south facing monocrystline systems)
- SEG report:
 - modeled as a three tiers; 5p(low), 12P(average) without battery, 29P peak flux octopus energy with battery
- Self consumption rate: 50% without battery, 70% with battery (energy saving trust guidance)

important limitations

This study assumes the household in question use and are actively charging an ev using the mandatory charger, based on UK annual mileage (7,400 miles/yr) and average EV efficiency (3.5 miles/Kwh) , given an assumed EV electrical demand of 2114kWh/yr. Where a buyer does not own and electric vehicle, total household electricity demand falls a subsequent 2114KWh/yr. which would proportionally increase the percentage of demand met by PV panels and modestly improve self self consumption financial benefit figures A sensitivity scenario for non EV owning household is recommended as further work but is outside the scope of this analysis given EV chargers are a mandatory fixture regardless of weather it is used or not.

0.3 PV System Sizing and generation

Panels counts were based of an average number of south facing panels feasible per roof taken from Blindwells site data rounded to a whole panel count.

House Type	Panels	System Size (kWp)	Annual Generation (kWh)
Kinloch	4	1.34	1125
Rosedale	6	2.01	1688
Pinehurst	7	2.35	1974
Sunningdale	8	2.68	2251

Table 1: PV system sizes of differing house types

annual generation = system size irradiance x performance ration.

0.4 Total Household demand

Total household demand combines Sap modeled delivered energy(space heating, hot water, lighting., appliance,cooking) with assumed EV charging demand

House	SAP Demand	EV Demand	Total Demand	%Met by PV
Kinloch	2060	2114	4174	27
Rosedale	2650	2114	4764	35.4
Pinehurst	2812	2114	4926	40.1
Sunningdale	3118	2114	5232	43

Table 2: Total Demands

these figures show that even before considering self consumption limitations, PV could offset over a 25% of total demand at the smallest house type rising to over 40% at the largest.

0.5 Three Scenarios

three scenarios were modeled to show the buyers financial benefit as more technology is added to the baseline PV installation.

0.5.1 Scenario 1- PV Only (no battery, 50 self consumed)

With no battery storage, PV use and instillation by itself allows for a improvement within energy exported and self consumption rate. a realistic self consumption rate for this situation is one of 50%.

House	Annual Benefit (£)
Kinloch	204.80
Rosedale	307.30
Pinehurst	323.20
Sunningdale	409.62

Table 3: PV only annual benefit

0.5.2 Scenario 2 - PV + Smart Cylinder Diverter (no battery hardware cost)

A smart diverter routes midday PV surplus directly to hot water cylinder immersion heater. This reduces the load put onto the ASHP for domestic hot water. This is assumed to eliminator $\approx 60\%$ of the pumps electric loads, at a significantly lower cost to a traditional electric battery.

House	Extra Savings (£)	Total annual benefit (£)
Kinloch	25.89	230.69
Rosedale	33.31	340.61
Pinehurst	38.38	361.58
Sunningdale	45.65	455.27

Table 4: annual Extra benefit vs Scenario 1

0.5.3 Scenario 3 - PV + 5KWH Battery (70 % self-consumed, octopus energy 29p export)

A in in house battery is capable of increasing the self consumption to $\approx 70\%$ and enables to enrollment ton the octopus intelligent flux tariff, where energy can be exported at up to 29p/kWh during peak demand rather than the 12p average rate.

House	Annual Benefit (£)
Kinloch	297.60
Rosedale	435.17
Pinehurst	508.77
Sunningdale	580.26

Table 5: Scenario 3 benefits

0.6 Instillation Cost and Payback(placeholder for price)

House	Scenario	Cost	Static Price	Payuback w7%
Kinloch	Pv only		9.81	7.97
	+cylinder		12.6	9.58
	+Battery		24.9	15.66
rosedale	PV Only		9.81	7.97
	+cylinder		11.49	8.84
	+Battery		19.34	15.66
Pinehurst	PV Only		10.9	8.78
	+cylinder		12.24	9.15
	+Battery		17.54	12.55
Sunningdale	PV Only		9.81	7.97
	+cylinder		10.8	7.9
	+Battery		16.23	11.85

Table 6: summary of PV upgrade scenarios

a 7% annual electricity price growth rate (UK historic average) was applied as a sensitivity check, since rising prices increase the value of both self consumed and exported electricity over time. Taking this growth into account it meaningfully shortens all the payback periods across all three scenarios, with the largest proportional improvement seen in the battery scenario. <https://www.overleaf.com/project/6a327c634a18480e60130>

0.7 Property value uplift

other than on going running cost savings, PV may add capital value at the point of resale. A 2024 Swansea university study, using ≈ 5 million property listings cross reference against UK Land Registry transaction data/ This found that PV panels can increase a property value by 6.1-7.1% . This uplift only applies to owned PV systems rather than leased systems as this can complicate resale and may deter buyers.

House	Sale Price (Blindwells)	6.1-7.1%
Kinloch	290,000	18,239-21,229
Rosedale	353,995	21,594-25,134
Pinehurst	412,495	25,162-29,287
Sunningdale	530,000	32,330-37,630

Table 7: Uplift value from previous sales

Given the houses already have strong EPC ratings from the ASHP and fabric performance alone, the uplift on property value from PV in this case is likely driven more by buyer perception of a green/modern home.though from 2025 onward this may unlock green mortgage rates fro buyers, which is further indirect benefit worth noting to Bellways sales teams

0.8 Conclusion and Recommendation

- Even without battery,PV delivers a meaningful annual saving and allows for a large resale value uplift, likely outweighing install cost many times over.
- The smart cylinder diverted offers the best cost to benefit ration of the three scenario. It cheap to install, has a meaningful improved payback and provides no need for the expense of a battery.
- The battery delivers the highest absolute annual benefit and the most dramatic improvement under ring electricity prices , but has the weakest static pay back rate out of the 3 scenarios. Its in the best position for a premium up sell rather than a standard spec.
- Recommend Bellway consider a PV+ smart cylinder as a standard "green value" package on ASHP sites, with a battery offered as up an optional premium upgrade.

Bibliography

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- [11] Swansea University. The Effect of Solar Panels on UK Residential Property Values. 2024. Based on approximately five million UK property listings cross-referenced with Land Registry transaction data.

.1 Appendix A

.1.1 Annual PV Generation

Formula:

Annual Output (kWh) = System Size (kWp) $\times$ Irradiance (kWh/m²/yr) $\times$ Performance Ratio

Assumptions: Irradiance = 1,050 kWh/m²/yr; Performance Ratio = 0.80

Table 8: Annual PV Generation by Property

House	System Size	Calculation	Annual Generation
Kinloch	1.34 kWp	$1.34 \times 1,050 \times 0.80$	1,125 kWh/yr
Rosedale	2.01 kWp	$2.01 \times 1,050 \times 0.80$	1,688 kWh/yr
Pinehurst	2.35 kWp	$2.35 \times 1,050 \times 0.80$	1,974 kWh/yr
Sunningdale	2.68 kWp	$2.68 \times 1,050 \times 0.80$	2,251 kWh/yr

.1.2 ASHP Electricity Consumption

Formula:

$$\text{ASHP Electricity (kWh/yr)} = \frac{\text{Heat Demand (kWh/yr)}}{\text{COP}}$$

COP values taken at A2/W30 from Grant Logic Air Monoblock datasheet [6].

.1.3 EV Charging Demand

Formula:

$$\text{EV Demand (kWh/yr)} = \frac{\text{Annual Mileage (miles)}}{\text{EV Efficiency (miles/kWh)}}$$

Table 9: ASHP Electricity Consumption by Property

House	Heat Demand	COP	ASHP Electricity
Kinloch	2,060 kWh	4.66	442 kWh/yr
Rosedale	2,650 kWh	4.66	569 kWh/yr
Pinehurst	2,812 kWh	4.29	655 kWh/yr
Sunningdale	3,118 kWh	4.00	780 kWh/yr

$$= \frac{7,400}{3.5} = 2,114 \text{ kWh/yr}$$

Source: Department for Transport National Travel Survey [14]; average EV efficiency based on typical UK passenger EV fleet.

.1.4 Total Household Electricity Demand

Formula:

$$\text{Total Demand} = \text{SAP Delivered Energy} + \text{EV Demand}$$

SAP Delivered Energy (DDER) sourced from AES Keltly modelling data [2].

Table 10: Total Household Electricity Demand

House	SAP Demand	EV Demand	Total Demand
Kinloch	2,060 kWh	2,114 kWh	4,174 kWh
Rosedale	2,650 kWh	2,114 kWh	4,764 kWh
Pinehurst	2,812 kWh	2,114 kWh	4,926 kWh
Sunningdale	3,118 kWh	2,114 kWh	5,232 kWh

.1.5 Self-Consumption and Export Split

Without battery (50% self-consumed, 50% exported) [12]:

$$\text{Self Consumed} = \text{PV Generation} \times 0.50$$

$$\text{Exported} = \text{PV Generation} \times 0.50$$

With battery (70% self-consumed, 30% exported) [12]:

$$\text{Self Consumed} = \text{PV Generation} \times 0.70$$

$$\text{Exported} = \text{PV Generation} \times 0.30$$

.1.6 Annual Financial Benefit

Formula:

$$\text{Total Annual Benefit} = (\text{Self Consumed} \times 24.4\text{p}) + (\text{Exported} \times \text{SEG Rate})$$

Worked example — Sunningdale, with battery, Octopus Flux (29p):

$$\begin{aligned} &= (1,576 \times £0.244) + (675 \times £0.29) \\ &= £384.54 + £195.75 = £580.29/\text{yr} \end{aligned}$$

.1.7 Smart Cylinder Diverter Saving

Hot water assumed at 40% of total annual heating load. Smart diverter eliminates ~60% of this load by using free PV surplus instead of grid electricity.

Formula:

$$\text{Extra Saving} = \text{ASHP Electricity} \times 0.40 \times 0.60 \times £0.244$$

Worked example — Sunningdale:

$$= 780 \times 0.40 \times 0.60 \times £0.244 = £45.65/\text{yr}$$

.1.8 Payback Period — Static

Formula:

$$\text{Payback (yrs)} = \frac{\text{Install Cost (£)}}{\text{Annual Benefit (£/yr)}}$$

.1.9 Payback Period — With Electricity Price Growth

Formula (compound growth, interpolated):

$$\text{Cumulative Savings at Year } N = \sum_{n=1}^N \text{Year 1 Benefit} \times (1.07)^{n-1}$$

Payback occurs when cumulative savings $\geq$ install cost. Exact year interpolated as:

$$\text{Payback} = Y - 1 + \frac{\text{Install Cost} - C_{Y-1}}{B_Y}$$

Where:

- Y = first year cumulative savings exceed install cost
- C_{Y-1} = cumulative savings at end of previous year
- B_Y = benefit in year Y

Growth rate of 7%/yr used, reflecting UK historic average electricity price inflation.

Full SAP Calculation Printout					
Plot Reference	4907-Q919-9082-109		Issued on Date		09/06/2026
Assessment Reference	Plot 109	Plot Type Ref	Pinehurst DET (CP)		
Plot Address	Plot 109, Pinehurst DET (CP)		SAP Version	10.2	
SAP Rating	85 B	DER	0.27	TERR	4.00
Environmental	07 A	% DER < TERR	N/A		
CO ₂ Emissions (t/year)	0.45	DDER	01.87	TDDER	27.03
Compliance Check	See Compliance Report		% DDER < TDDER	18.72	
Dwelling Space Heat Demand	34.68Whm ² /yr		Dwelling Space Heat Demand Limit		36.01Whm ² /yr
Assessor Details	Mr. Michael Black		Assessor ID		PE176-0001
Client	Bellway Homes, Scotland East				

Figure 1: Enter Caption

Full SAP Calculation Printout						
Plot Reference	4907-Q919-9082-043			Issued on Date	09/06/2026	
Assessment Reference	Plot 043	Plot Type Ref	Rosedale DET (AS)			
Plot Address	Plot 043, Rosedale DET (AS)			SAP Version	10.2	
SAP Rating	03 B	DER	0.91	TERR	4.41	
Environmental	06 A	% DER < TERR	N/A			
CO ₂ Emissions (t/year)	0.42	DDER	26.24	TDDER	29.91	
Compliance Check	See Compliance Report		% DDER < TDDER	12.27		
Dwelling Space Heat Demand	37.27Whm ² /yr	Dwelling Space Heat Demand Limit		37.36Whm ² /yr		
Assessor Details	Mr. Michael Black			Assessor ID	PE176-0001	
Client	Bellway Homes, Scotland East					

Figure 2: Enter Caption

.2 Property Value Uplift

Formula:

$$\text{Value Uplift} = \text{Sale Price} \times \text{Uplift Rate}$$

Uplift rates sourced from Swansea University (2024) [15]: 6.1–7.1% (academic, Land Registry transaction data); conservative industry estimate: 1–4%.

Worked example — Sunningdale (6.1–7.1%):

$$= £530,000 \times 0.061 = £32,330$$

$$= £530,000 \times 0.071 = £37,630$$

.3 SAP Values Used

Full SAP Calculation Printout					
Plot Reference	4907-Q919-9082-076	Issued on Date		09/06/2026	
Assessment Reference	Plot 076	Plot Type Ref		Kilninch SEM (CP)	
Plot Address	Plot 076, Kilninch SEM (CP)	SAP Version		10.2	
SAP Rating	85 B	DER	3.94	TERR	4.19
Environmental	07 A	% DER < TERR		N/A	
CO ₂ Emissions (t/year)	0.33	DDER	23.96	TDDER	28.87
Compliance Check	See Compliance Report	% DDER < TDDER		16.17	
Dwelling Space Heat Demand	28.42Whm ² /yr	Dwelling Space Heat Demand Limit		29.51Whm ² /yr	
Assessor Details	Mr. Michael Black	Assessor ID		PE176-0001	

Figure 3: Enter Caption

Full SAP Calculation Printout						
Plot Reference	4307-QP19-0002-108			Issued on Date	09/06/2020	
Assessment Reference	Plot 108		Plot Type Ref	Sunningdale DET (DP)		
Plot Address	Plot 108, Sunningdale DET (DP)			SAP Version	10.2	
SAP Rating	SB B	DER	2.78	TER	3.87	
Environmental	07 A	% DER < TER	N/A			
CO ₂ Emissions (g/kWh)	0.4	% DER	10.83	TER	25.02	
Compliance Check	See Compliance Report		% DER < TER	28.25		
Dwelling Space Heat Demand	15.37kWh/m ² /yr		Dwelling Space Heat Demand Limit	18.09kWh/m ² /yr		
Assessor Details	Mr. Michael Stark			Assessor ID	19079-0001	
Client	Railway Homes, Sunningdale East					

Figure 4: Enter Caption